

How BloodLink works, and how to test all of it

A blood-donation platform built around registered **blood banks**. Four roles, one spine: a recipient asks for blood, the system scores and ranks compatible donors by availability and distance, an officer matches one and records the donation into their bank's stock, and everyone locates and routes to the nearest bank. This manual walks each role end to end, maps how the pieces connect, explains the enhancements that make it "enhanced," and gives you a role-by-role QA script.

01 / The cast

Four roles, at a glance

Everyone signs in through one of two doors — the admin portal (`login.php`) or the shared user portal (`user-login.php` , role dropdown). The session `role` gates every page.

SESSION ROLE: ADMIN

Administrator

Runs the bank network: manages blood banks, donors, hospital officers, recipients, and per-bank stock; posts emergency alerts; monitors the request queue; searches the donor pool; reads charts and reports.

SESSION ROLE: OFFICER

Hospital Officer

Belongs to a blood bank. Works the request queue and the Emergency Board, matches each request to the best-scored available donor, and **confirms the donation** — which banks the units into their own bank's stock.

SESSION ROLE: DONOR

Blood Donor

Registers with a blood group and address, sets their **availability** (status, schedule, contact preference), checks eligibility (56-day rule), sees emergency alerts, and routes to the nearest bank.

SESSION ROLE: RECIPIENT

Recipient

Registers, files blood requests (urgency + how soon the blood is needed), tracks each request's status and any donation received, and uses the locator.

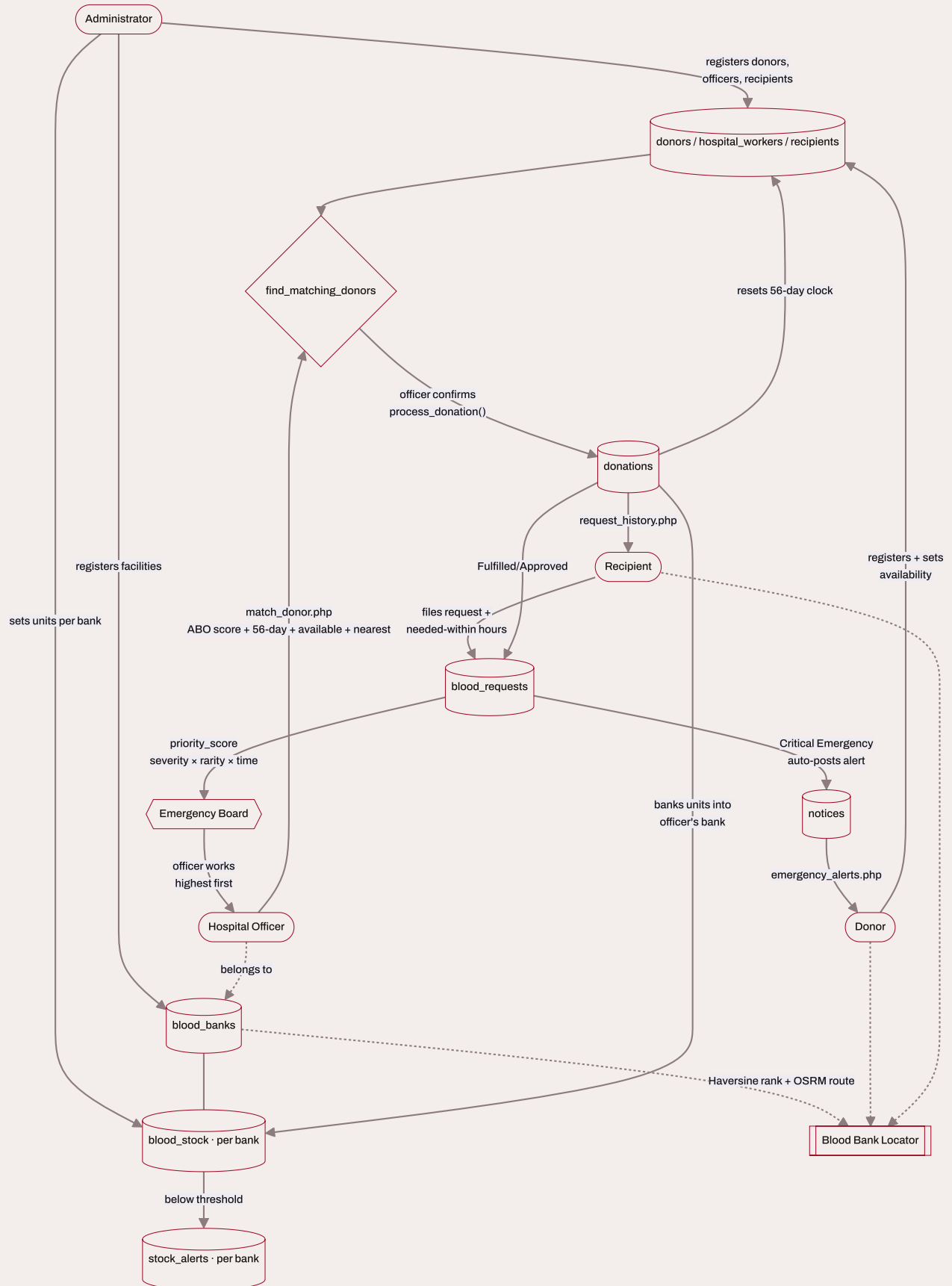
WHAT MAKES IT "ENHANCED"

Over the base blood-bank workflow, BloodLink adds four things a real blood service needs: a **donor availability registry** with a Find Donors search, an **emergency workflow** that scores and ranks requests by priority with live countdowns, **compatibility-aware matching** (ABO/Rh score + blood-type rarity, availability-first), and **analytics dashboards**. Section 05 covers all four. The codebase is now purely a blood-donation app — every table, role and label speaks that vocabulary; there are no leftover names from earlier work.

02 / The spine

How the whole app connects

Everything orbits one workflow: a request becomes a matched, recorded donation banked at a specific facility. The locator, the 56-day rule, availability and per-bank stock all hang off it.



- **Admins** register the `blood_banks` (facilities), the people, and each bank's `blood_stock`.
- **Recipients** create `blood_requests` with an urgency and (optionally) how soon the blood is needed; the app computes a `priority_score`. A Critical Emergency also writes a donor-facing `notices` alert.
- **Officers** belong to a bank. The **Emergency Board** ranks open requests by priority; `match_donor.php` runs `find_matching_donors()` ($\text{ABO/Rh score} \times 56\text{-day} \times \text{availability} \times \text{nearest}$); `process_donation()` writes the `donations` row in one transaction that resets the donor's clock, **banks the units into the officer's bank**, updates the request, and clears that bank's alert.
- **Donors & recipients** use the **Blood Bank Locator**: it ranks banks nearest-first and draws an OSRM driving route to the chosen one.

Ten tables, and who touches them

TABLE	HOLDS	WRITTEN BY	READ BY
login	Admin accounts	seed	admin login
blood_banks	Facilities (name, address, geocoded lat/lng, contact)	admin (manage banks)	locator, stock, officers, reports
hospital_workers	Officers (+ blood_bank_id, position)	admin (add officer → picks a bank)	officer login, matching, reports
donors	Donors (+ blood_group, last_donation_date, lat/lng, is_available , schedule , contact pref)	donor reg / self-service, admin add donor	matching, Find Donors, map, reports
recipients	Patients (+ bcrypt password, lat/lng)	recipient reg, admin add recipient	recipient login, requests
blood_requests	Requests (+ urgency, severity , required_within_hours , priority_score , status, lat/lng)	recipient (request_blood)	queue, Emergency Board, match, history
donations	Donation ledger (+ blood_bank_id)	officer (process_donation)	histories, reports
blood_stock	Units per bank per blood type	admin, process_donation	stock page, dashboard, locator, reports
stock_alerts	Low-stock alerts per bank per type	stock page, process_donation	dashboard, stock page
notices	Emergency alerts	officer/admin, auto (critical request)	donor emergency alerts

This matches the brief's ERD (Figure 5.3): the design centres on `blood_banks`, with `blood_stock` and `stock_alerts` scoped per bank and `hospital_workers` belonging to a bank. The donor and request tables carry the extra columns behind the availability and emergency-priority features.

What each role does, end to end

ADMINISTRATOR runs the network

Signs in at `login.php` (`admin` / `admin`) → `dashboard.php` .

- **Blood Banks:** add / edit / delete facilities — each address is geocoded onto the map, and a new bank gets an empty stock row per blood type automatically.
- **People:** Manage Donors, Hospital Officers (each assigned to a bank), Recipients — full CRUD; officer & recipient passwords hashed on create.
- **Blood Stock:** pick a bank, then set units + low-stock threshold per type; dropping below threshold raises that bank's alert, restocking resolves it.
- **Emergency Board & Find Donors:** see every open request ranked by priority with live countdowns, and search the donor pool by group, compatibility and availability.
- **Oversight:** the Blood Requests queue (read-only for admin), Emergency Alerts, the Geo-Map/locator, and Reports — now with a blood-group chart, a donation-status doughnut and a stock-by-group chart, plus CSV export.

HOSPITAL OFFICER makes the match

Signs in at `user-login.php` (Hospital Officer, email + password) → `officer_dashboard.php` . Each officer belongs to one bank.

- **Emergency Board:** open requests ranked by `priority_score` , each with a live countdown to its deadline. Work the top of the list first.
- **Blood Requests & Find Donors:** the open queue, and a search over the donor pool (available-first, best compatibility, nearest).
- **Match & confirm:** `match_donor.php` lists ABO/Rh-compatible, 56-day-eligible donors with a **Match score** and an **Availability** flag, ranked available-first then by score then nearest. Confirming runs `process_donation()` , which banks the units into *this officer's bank*.
- **Emergency Alerts & Geo-Map.**

is the supply

Registers at `register.php` (blood group + geocoded address), signs in with Donor ID + password → `donor_dashboard.php`.

- **My Availability:** set status (available or not), a schedule (anytime, weekends, evenings, emergencies...), and a contact preference. Available donors surface first when an officer searches.
- **Donate Blood:** shows eligibility (inside/outside the 56-day window). Officers record the actual donation on match.
- **Emergency Alerts and the Blood Bank Locator** (Nearby Banks).

creates the demand

Registers, signs in with email + password → `recipient_dashboard.php`.

- **Request Blood:** patient, group, units, hospital, urgency, and optionally *needed within (hours)*. The app derives a severity and a priority score; Critical Emergency auto-posts an alert; one pending request per group is enforced; the hospital is geocoded.
- **Request History** (status + donations received) and the **Locator**.

Four features that lift it above a basic register

These are the additions that make BloodLink an *enhanced* system — each ties back to the spine in Section 02.

01 • AVAILABILITY REGISTRY

Donors say when they can give

Each donor keeps an `is_available` flag, a schedule and a contact preference, edited from their own dashboard. The admin/officer **Find Donors** page searches the pool by group and mode (compatible / exact group), optionally filtered to available donors and ranked from a chosen bank.

02 • EMERGENCY WORKFLOW

Requests are triaged, not just queued

A request carries a severity and an optional deadline. The **Emergency Board** orders open requests by `priority_score` and shows a live countdown per request, so the most urgent, rarest, soonest-needed case is always at the top.

03 • COMPATIBILITY-AWARE MATCHING

The system scores every donor

Matching goes beyond "same group": a compatibility score rewards exact matches and universal O-, blood-type rarity breaks ties, and available donors sort first. The officer sees the score and availability right in the match table.

04 • ANALYTICS

At-a-glance charts

The admin dashboard charts current stock by blood group; Reports adds a donor blood-group distribution and a donation-status doughnut (Completed / Pending / Cancelled), alongside the existing KPIs and CSV export.

HOW THE PRIORITY SCORE IS COMPUTED

```
priority = severity (critical 60 · severe 40 · moderate 20 · low 5)
          + rarity × 2 (AB- rarest 10 ... 0+ most common 3)
          + time pressure (≤3h 20 · ≤12h 12 · ≤48h 6 · else 2)
          → capped at 100
```

A Critical, AB-, needed-within-2-hours request therefore scores $60 + 20 + 20 = 100$ and leads the board. The compatibility score is 0 if incompatible, else 100 (exact) / 90 (donor O-) / 80 (same ABO) / 70 (otherwise compatible).

06 / Blood Bank Locator

The interactive map (brief Fig 5.5)

The locator answers "which registered bank is nearest, and how do I get there?" — proximity only; stock is managed separately.

1

Capture location

"Use my location" (browser GPS) or "Use Kaduna city centre". A blue marker shows where you are.

2

Rank the banks

Each bank's distance is computed with the Haversine formula and the list is sorted nearest-first, with a colored distance badge (green ≤ 5 km, amber 5–15, red >15) and dashed connectors to the three nearest.

3

Route to one

Click a bank's "Directions": OSRM returns a driving route drawn as a polyline **colored by distance band**

, and the route box shows

distance by road + estimated drive time

.

Built entirely on free, open services — Leaflet + OpenStreetMap tiles, Nominatim geocoding, OSRM routing — so there is no API key or billing account, unlike Google Maps.

07 / The end-to-end donation

Follow one unit of blood

1

Recipient files a Critical O- request

needed within a few hours → a `blood_requests` row (Pending) with a high `priority_score` + an auto `notices` alert.

2

Donors see the alert

on their Emergency Alerts page; available O- donors are ready in the pool.

3

Officer opens the Emergency Board

the request sits at the top by priority, its countdown running → Match Donor.

4

System proposes donors

only O-, 56-day-eligible,

available first

, each with a compatibility score and distance to the hospital.

5

Officer confirms

process_donation(): writes `donations` (with the officer's `blood_bank_id`), resets the donor's clock, adds units to that bank's `blood_stock`, flips the request, resolves that bank's alert.

6

Everyone sees the result

Recipient history shows Fulfilled; the donor is inside a fresh 56-day window; the bank's stock, the dashboard charts and Reports update.

08 / QA runbook

How to test the whole app

Set up once, then run each role's script. The seed is built for this — 5 real banks, donors on the 56-day boundary, all eight blood groups, two donors marked unavailable, and requests pre-scored for the Emergency Board.

SETUP

```
mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS donor_app"
mysql -u root -p donor_app < db/schema.sql
mysql -u root -p donor_app < db/seed.sql
# app under Local: http://<site>.local/bloodlink/ · re-run schema.sql + seed.sql
```

ROLE	PAGE	USERNAME	PASSWORD
Admin	login.php	admin	admin
Officer	user-login.php	maryam@bloodlink.test	Officer@123
Donor	user-login.php	BL-KD-001	Donor@123
Recipient	user-login.php	hassan@bloodlink.test	Recipient@123

A. AUTHENTICATION & ACCESS

- ✓ All 4 logins land on the right dashboard; wrong passwords rejected.
- ✓ Logged out, a protected page redirects (302); a donor can't reach an admin/officer URL.
- ✓ Logout works from the topbar avatar dropdown and the sidebar, every role.

B. ADMINISTRATOR

- ✓ **Blood Banks:** add a bank (address geocoded), edit it, delete it; a new bank shows 8 stock rows.
- ✓ Add a donor (real blood type shows, not "N/A"), an officer (assigned to a bank), a recipient — each new officer/recipient can then log in.
- ✓ **Blood Stock:** choose a bank; drop A+ below threshold → that bank's alert appears + dashboard count rises; save again → no duplicate; restock → resolves.
- ✓ **Charts:** dashboard shows the stock-by-group bar; Reports shows the blood-group bar + status doughnut; Total Donations equals the donations count; CSV export downloads.
- ✓ **Find Donors:** search O- (compatible) with "available only" → available O- donors listed with a match score.

C. RECIPIENT → EMERGENCY BOARD → OFFICER → DONOR (THE SPINE)

1

Recipient files O-, Critical Emergency, needed within 3 hours

- ✓ Shows Pending in Request History; a matching alert exists; the request gets a high priority score; duplicate/invalid submissions refused.

2

Officer opens the Emergency Board

- ✓ The new request ranks at/near the top by score; its countdown ticks down; lower-priority requests sit below.

3

Officer matches

- ✓ Only O- donors; available-first, each with a match score and distance; day-30 donor excluded, day-56 included.

4

Officer confirms

- ✓ Request → Fulfilled; the officer's bank's O- stock changes; the donation row carries that bank_id; re-confirming the same donor is refused.

D. DONOR AVAILABILITY

- ✓ On the donor dashboard, toggle availability off, change schedule + contact preference, Update → confirmation shows and the values persist on reload.
- ✓ An unavailable donor drops below available ones in Find Donors (and off the list with "available only" ticked).

E. BLOOD BANK LOCATOR

- ✓ "Use Kaduna city centre" ranks banks nearest-first with colored distance badges.
- ✓ Click "Directions" → a colored road route draws and the box shows distance + drive time.
- ✓ OpenStreetMap attribution is visible; banks show units in stock in their popup.

F. CROSS-CUTTING UI/UX

- ✓ No content sits under the fixed topbar/sidebar; tables scroll rather than stretch.
- ✓ Consistent chrome across all logged-in pages, all four roles.
- ✓ Accented text renders correctly (e.g. "Officer – Hospital"); phone width collapses the sidebar to a hamburger.
- ✓ A tampered/expired form token is rejected (CSRF).

KNOWN LIMITATIONS (EXPECTED)

- **Nominatim rate limit (1/sec):** registering banks/people rapidly may store NULL coordinates — they won't appear on the map until re-saved.
- **OSRM demo server** is fine for a demo, not production traffic; self-host it for real deployment.
- **Admins can't confirm donations** by design — the officer is the clinical actor, and the donation's bank comes from the officer's bank.
- **Rotate the old Google Maps key** in the Google console — it's out of the code but in earlier git history.

BloodLink Operations & QA Manual (v4) · roles: admin / officer / donor / recipient · spine: blood_requests
→ priority_score → find_matching_donors → process_donation → donations (banked per facility) ·
enhancements: availability · emergency board · scored matching · analytics · locator on OpenStreetMap +
Nominatim + OSRM · 10 tables centred on blood_banks.